STANDALONE 3D ONLY

2 Points (Individual · Optional)

3D Graphics and Editor Basics

LEARNING OBJECTIVES

You learn the basics of the 3D editor in Godot: **MeshInstance3D**, **camera**, **lighting** and **environment**. You build a simple 3D scene with animated objects.

01 Build a 3D Scene

Create a new scene with `Node3D` as the root node. Add at least three `MeshInstance3D` nodes and assign a different `PrimitiveMesh` shape to each, for example `BoxMesh`, `SphereMesh` or `CylinderMesh`. Arrange the objects in the 3D viewport.

The 3D viewport works differently from the 2D viewport. Use the middle mouse button to rotate, scroll to zoom, and `Shift` + middle mouse button to pan.

02 Camera, Lighting and Environment

Add a `Camera3D` and position it so that the objects are visible in the running game. Add at least one light source, for example `DirectionalLight3D` or `OmniLight3D`, and adjust its colour and intensity. Also create a `WorldEnvironment` and experiment with the sky and background colour settings.

03 Animation

Add an `AnimationPlayer` and create an animation for at least one of the objects. Animate node properties such as `rotation`, `scale` or `position`. Set the animation to autoplay.

In 3D, Godot uses `Vector3` instead of `Vector2`. Rotation is specified in degrees or radians depending on the property.

RESOURCES

- Godot Docs: Introduction to 3D
- Godot Docs: Lights and Shadows
- Godot Docs: Environment and Post Processing

ACCEPTANCE REQUIREMENTS

- [ ] At least three 3D objects with different shapes are visible in the scene.
- [ ] A camera is present and the scene is visible in the running game.
- [ ] At least one light source and a background environment are set up.
- [ ] At least one object is animated in the running scene.